MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS)
THIRD SEMESTER EXAMINATION
(2024 ADMISSION ONWARDS)

MODEL QUESTION PAPER - THEORY

STATISTICS

MG3DSCSTA200 – STATISTICAL DISTRIBUTIONS

Duration: 2 Hours **Maximum: 70 Marks**

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short answer questions
Answer any ten questions. Each question carries **3 marks**.

1. Distinguish between discrete and continuous random variables. [U] [1]
2. Let X be a random variable with p.d.f $f(x)=\begin{cases}\left(\frac{1}{2}\right)^x & x=1,2,... \\ 0 & elsewhere\end{cases}$

   Find the p.d.f of $Y=X^2$.[An] [1]
3. Define MGF of a random variable. What is the MGF of geometric distribution? [K] [2]
4. Establish Additive Property of Poisson Distribution [A] [1,2]
5. Given that $f(x)=k\,x^4\,e^{-x}\ \ 0\le x\le\infty$ is a p.d.f. Determine k.[A] [1]
6. Find the m.g.f of X with p.d.f $f(x)=\frac{1}{\theta}$, $0<x<\theta$ and 0 elsewhere. [An] [2]
7. What are the properties of a normal distribution? [U] [1,4]
8. Obtain the characteristic function of $f(x)=\theta e^{-\theta x}\ \ 0<x<\infty$. [A] [2]
9. Write down the probability density function of Snedecor's F-distribution with ‘m’ and ‘n’ degrees of freedom). [K] [3]

10. Discuss the relationship between the t-distribution and the normal distribution. [A] [3]
11. Find the value of k such that P (X < k) = 0.95 for a Chi-square distribution with 3 degrees of freedom [An] [3]
12. Explain the relationship between the t-distribution and the χ^2 distribution [An] [3]

Part B

Short Essay questions

Answer any four questions. Each question carries **7 marks.**

13. For the following p.m.f find the distribution function and sketch its graph. [A] [1]

X	0	1	2	3
$f(x)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

14. A r.v X has the p.d.f $f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} - \infty < x < \infty$.Find the p.d.f of $Y = X^2$.[An] [1]
15. Derive the characteristic function of the binomial distribution and hence obtain its mean and variance [An] [2]
16. Define the exponential distribution and establish its lack of memory property. [A] [1,2]
17. Define a Chi-square variate with n d.f. and derive its distribution. [A] [3]
18. Describe the method to obtain the probability points of $F(n_2, n_1)$ from those of $F(n_1, n_2)$. [An] [3]

Part C

Essay questions

Answer any one question. Each question carries **12 marks**.

19. The following data is related to number of wrong calls received at a call centre during a year. Fit a Poisson distribution to this data and obtain the expected frequencies

x	0	1	2	3	4
Frequency	182	88	66	24	5

[An] [1,2]

20. State Lindeberg-Levy central limit theorem. Show that binomial distribution tends to normal distribution as $n \to \infty$[An] [1,4].